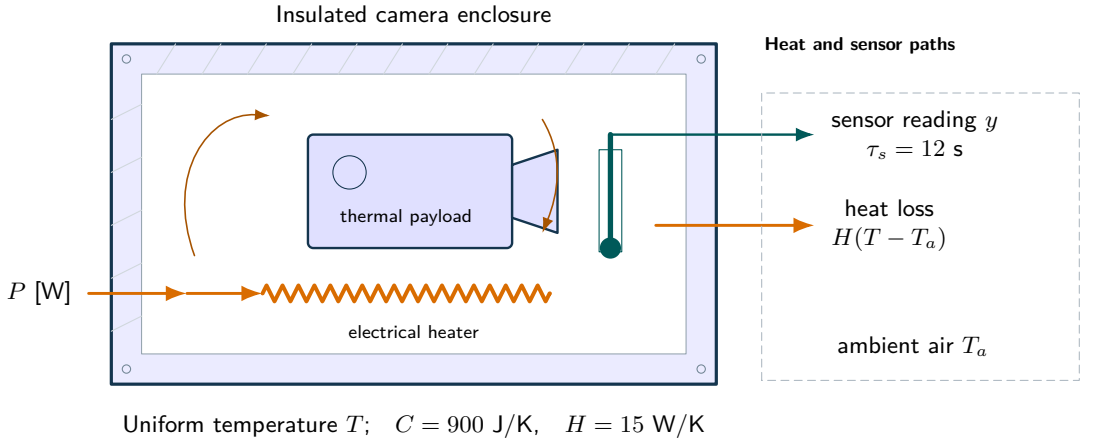
Physical systems and control models

Original vector illustrations for the hypothetical learning collection. Equipment views identify the modeled phenomenon; signal diagrams identify its mathematical structure. A picture does not establish experimental validation.

01. Thermal enclosure and a slow sensor



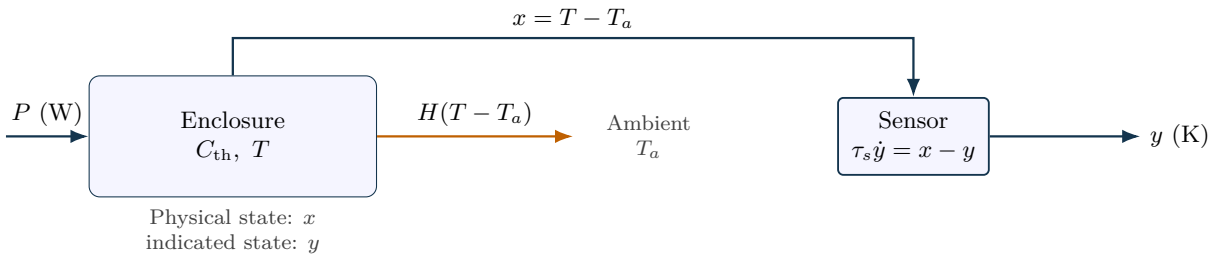
Stored and indicated heat
$x = T - T_a$ [K]; y [K].
Physical storage and sensor memory are separate states.

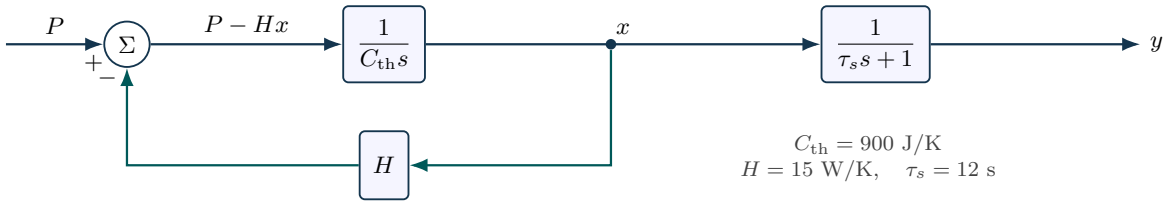
Balances
$900\dot{x} = P - 15x$.
$12\dot{y} = x - y$.
Heat rate in W; time in s.

Read the cutaway
Orange: heat flow. Teal: measurement. Wall and payload are one lumped capacity; no spatial field is resolved.

Cutaway of a lumped thermal enclosure. Orange arrows carry heat; the probe measures a lagged temperature. The drawing fixes the energy boundary, not a real device geometry.

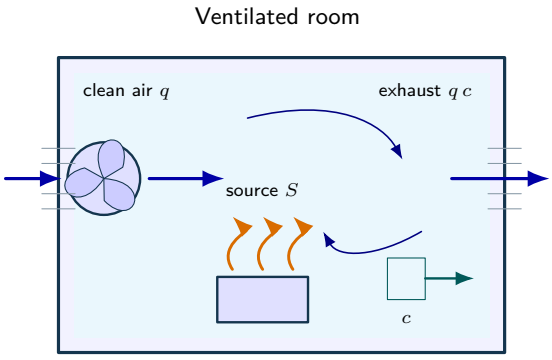
From equipment to equations



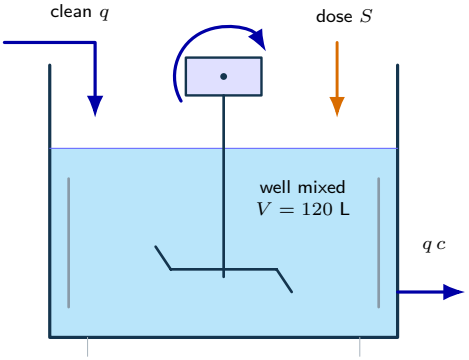


Thermal storage, heat leakage and sensor dynamics are separate elements. The lower loop implements the energy balance, not an added temperature controller.

02. Ventilation as a bilinear dynamical system



$V = 120$ m^3; concentration c



Liquid analogue: same balance, different units

Inventory and channels

c: tracer concentration.
S: source rate; q: throughput.
Equal inlet/outlet flow fixes V.

Balance and bilinearity

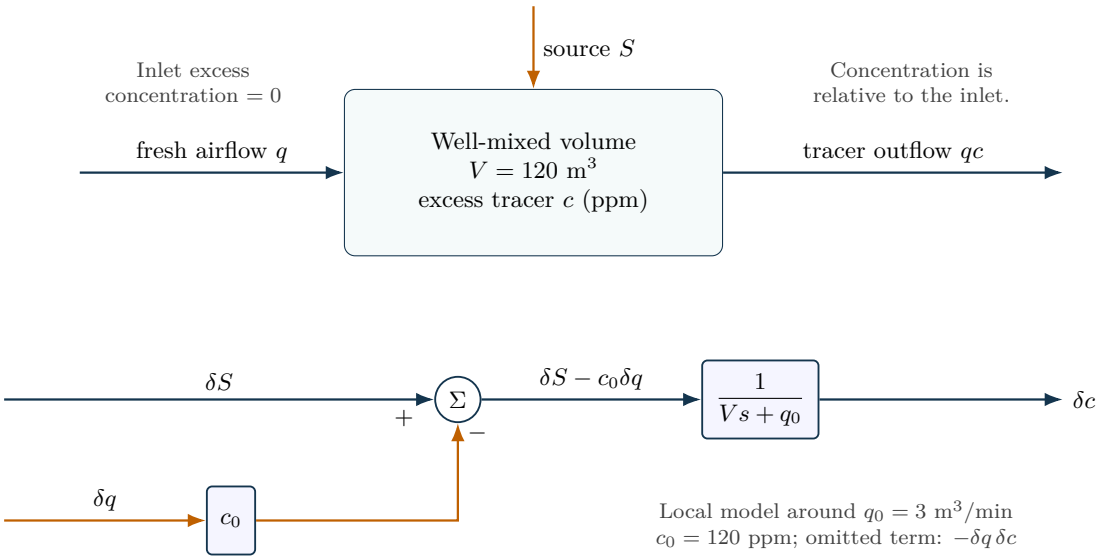
$V\dot{c} = S - qc$.
The product qc couples state and input. Room and liquid cases use different volume units.

Mixing assumption

Grilles and baffles illustrate ideal mixing only. Air velocity fields, fan dynamics and mixing transients are not modeled.

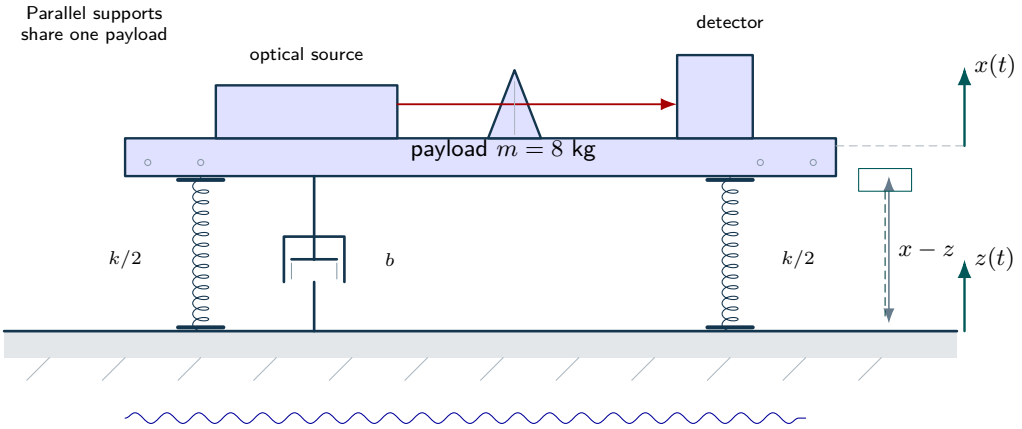
A ventilated room and a well-mixed liquid analogue share an inventory balance. The room uses m^3 and the liquid case uses L; do not mix their concentration or flow units. Equal inlet and outlet volume flows keep each inventory constant.

From equipment to equations



The physical balance and its local two-input signal model. Airflow enters with a negative sign because increased ventilation removes excess tracer.

03. A precision platform under floor motion



Measured coordinates

x, z: absolute displacements [m].
$x - z$: spring travel [m].
$\dot{x} - \dot{z}$: damping velocity.

Force balance

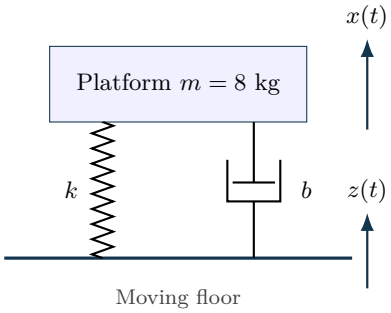
$m\ddot{x} + b(\dot{x} - \dot{z})$
$+k(x - z) = 0.$
Two $k/2$ supports give total k.

Operating boundary

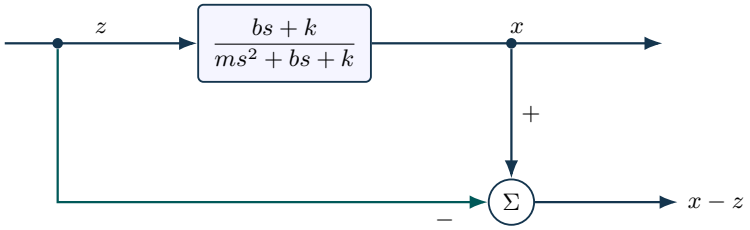
Coordinates are deviations about static sag. The blue floor trace indicates motion; it is not a measured waveform.

An optical payload rests on parallel spring and damping elements. The two springs each have stiffness $k/2$, giving total k. Both the floor and payload move; spring extension and damping velocity are relative quantities. Weight is balanced at the static equilibrium about which the model is written.

From equipment to equations



Moving floor
coordinates about loaded equilibrium

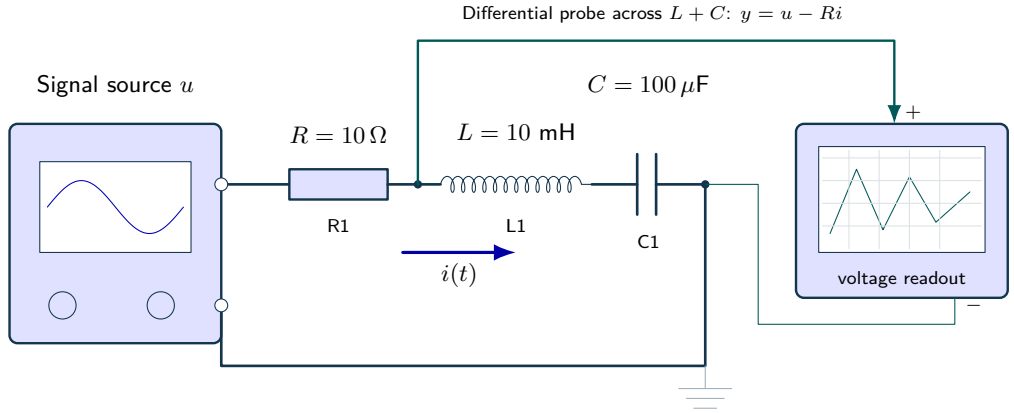


$$\omega_n = \sqrt{k/m}, \quad \zeta = \frac{b}{2\sqrt{mk}}$$

absolute motion and relative travel are different outputs

Base excitation drives absolute motion; subtracting the floor coordinate gives relative travel. The spring and damper act in parallel.

04. A passive notch filter and component tolerance



Ideal series network; the output port includes direct feedthrough

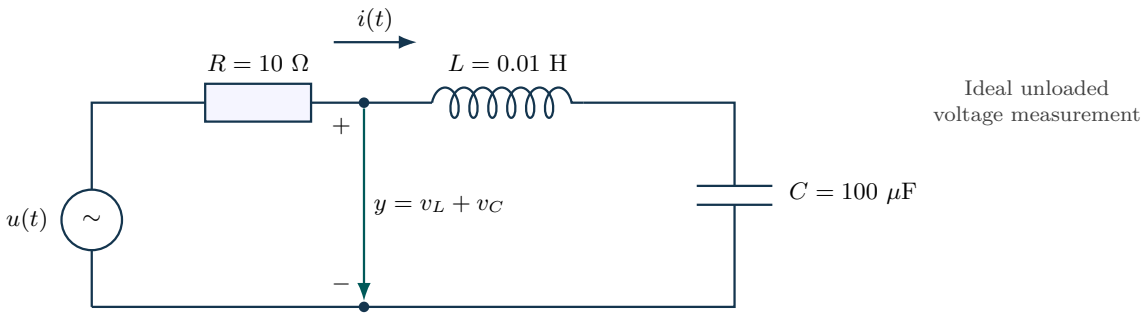
Electrical states
Current i [A] and capacitor voltage v_C [V] store magnetic and electric energy.

Circuit equations
$L\dot{i} = u - Ri - v_C$.
$C\dot{v}_C = i$.
Output: $y = u - Ri = v_L + v_C$.

Measurement boundary
The two probe leads span $L + C$. The depicted instrument display is illustrative, not a measured response.

A source excites the series RLC network; an ideal differential probe spans the combined inductor and capacitor. Measuring only the capacitor would produce a different transfer function. Wire placement is schematic.

From equipment to equations

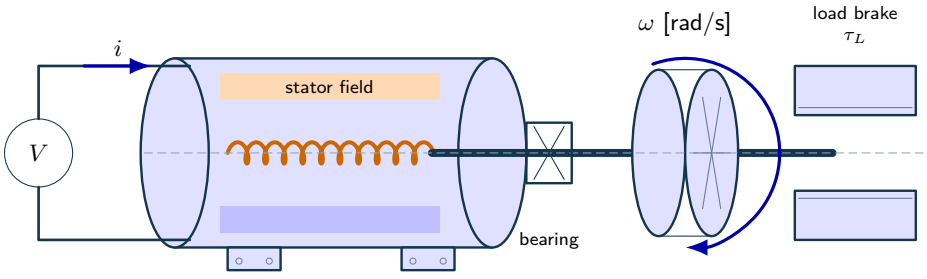


$$U(s) \longrightarrow \boxed{H(s) = \frac{LCs^2 + 1}{LCs^2 + RCs + 1}} \longrightarrow Y(s)$$

At $\omega_0 = 1/\sqrt{LC}$, reactive voltages cancel; current and stored energy need not vanish.

Series RLC network with output measured across L and C together. The output probe spans those two components, not the whole source.

05. Flywheel speed regulation under a load torque



Electrical channel
$L\dot{i} = V - Ri - k_e\omega$.
Back emf opposes the applied voltage; input V is in volts.

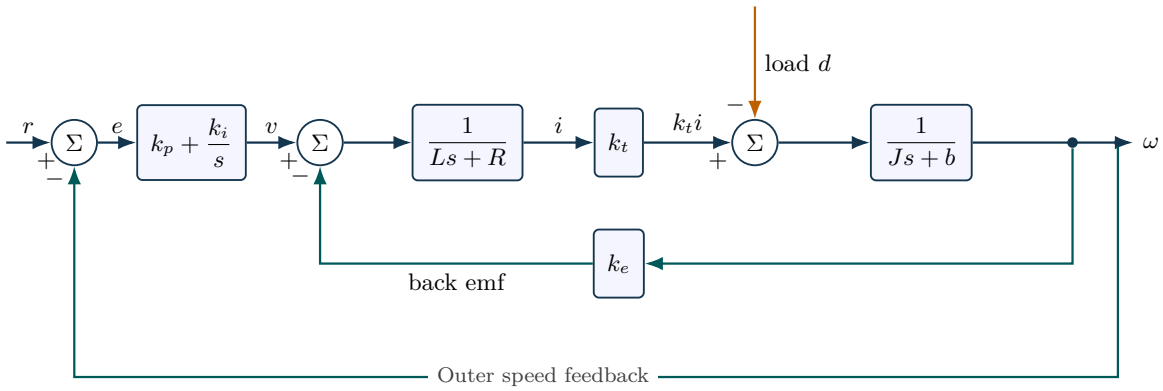
Mechanical channel
$J\dot{\omega} = k_t i - b\omega - \tau_L$.
The rotation arrow is centered on the flywheel axis. Positive load torque opposes positive speed.

Energy conversion
$k_t i$ is torque [N m]; ω is speed [rad/s]. Bearing and brake details add no unmodeled dynamic states.

A voltage-driven motor converts armature current into shaft torque. The rotor and external brake introduce inertia, friction and load torque. The illustrative cutaway does not add magnetic saturation or friction to the stated linear model.

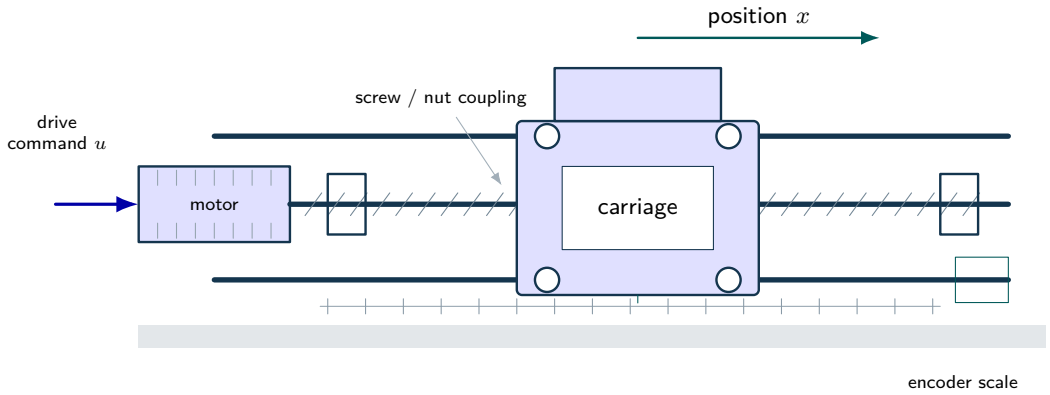
From equipment to equations

Electrical storage → torque conversion → mechanical storage
opposing load and back emf enter at different summing junctions



Outer PI speed regulation around the electromechanical motor. The inner back-emf path and opposing load torque have explicit negative signs.

06. Gain selection with a third-order actuator



Illustrative positioning axis; $G(s) = 1/[s(s+2)(s+6)]$ is stipulated

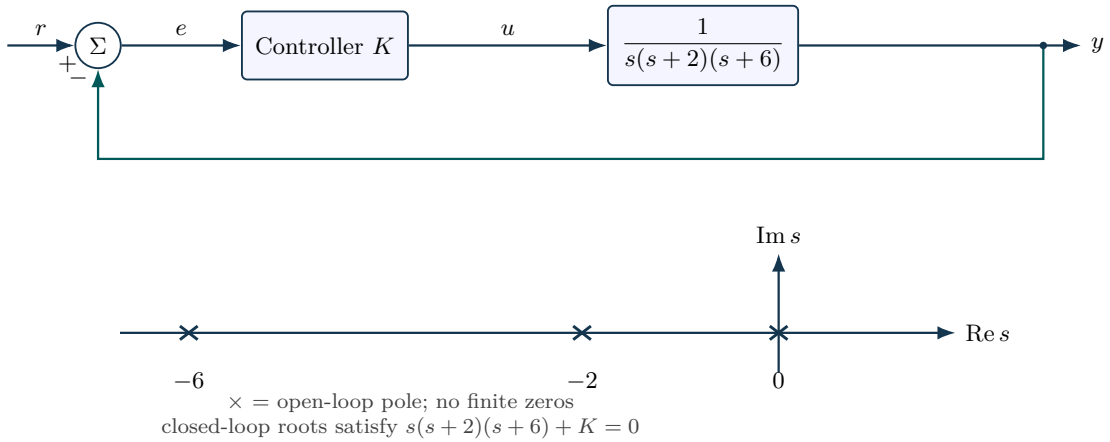
Command and readout
Drive command u and measured position x form a normalized teaching interface.

Stipulated dynamics
$G(s) = \dfrac{1}{s(s+2)(s+6)}$.
One integrator and two stable lags determine the pole geometry.

Geometry is illustrative
Screw pitch, motor constants, backlash and bearing friction are not inferred from the picture. Use the stated transfer function.

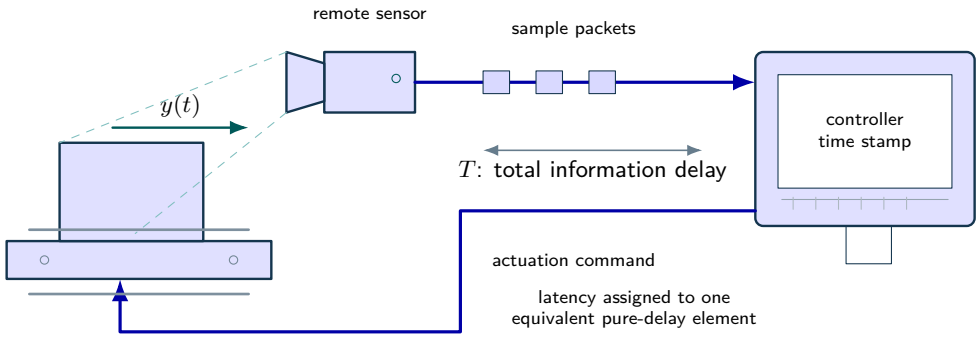
A motorized positioning axis illustrates the role of the integrator and two stable lags. The normalized transfer function is specified as a teaching model; the pictured dimensions do not identify its coefficients.

From equipment to equations



Unity negative feedback and the open-loop pole map. Pole symbols provide the starting geometry for the root-locus exercises.

07. A feedback loop with a communication delay



Mechanism and communication path; packet losses and jitter are omitted

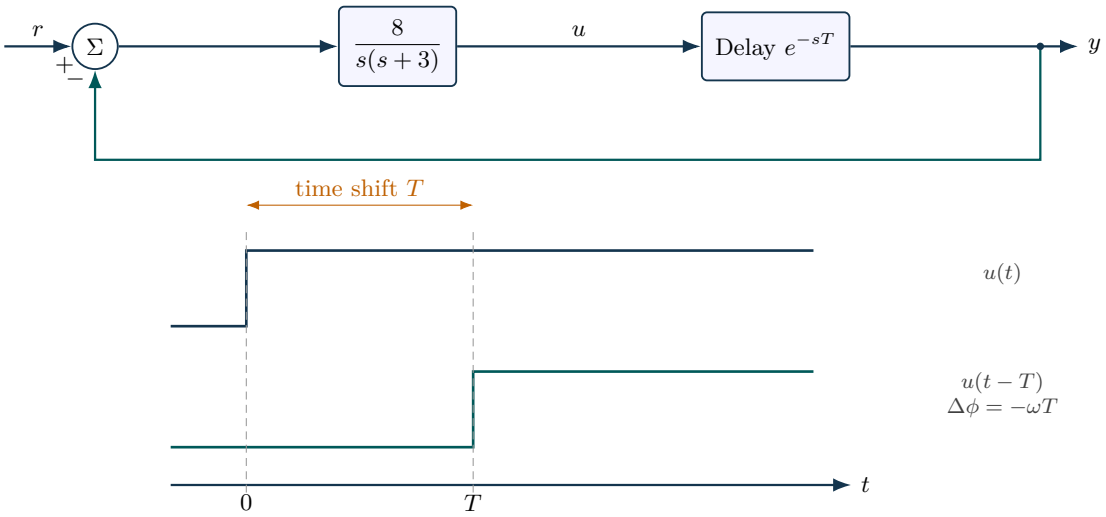
Signal direction
Top path: sensed output to controller. Bottom path: command returning to the mechanism.

Equivalent delay
$y_d(t) = y(t - T)$.
e^{-sT} has unit magnitude and phase $-\omega T$ [rad].

Assessment boundary
Fixed total latency is assumed. Packet jitter, dropouts and scheduling are excluded; packet boxes are conceptual.

A remote mechanism sends measurements and receives commands through a communication path. The model combines those latencies into one ideal constant delay; the drawn packet train is not an experimental record.

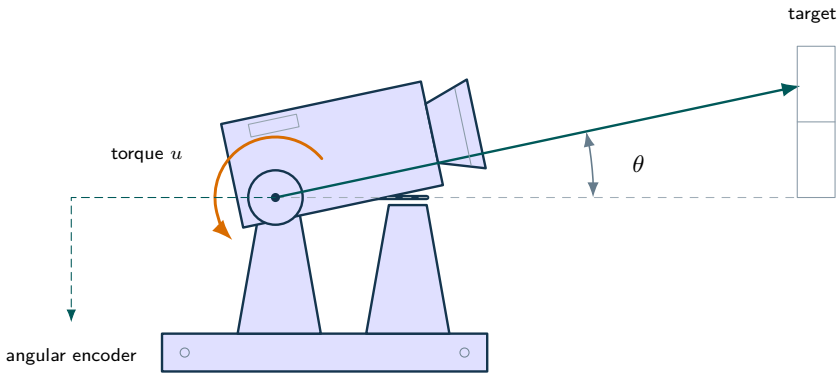
From equipment to equations



Equal-height unit steps, drawn on vertically offset tracks for clarity.

Constant pure delay shifts a signal in time and contributes frequency-dependent phase lag. The sketch shows ideal steps, not experimental records.

08. Lead-lag design for a normalized camera axis



Gimbal axis with angular readout

Normalized model $G(s) = 1/[s(s+1)]$; geometry is illustrative

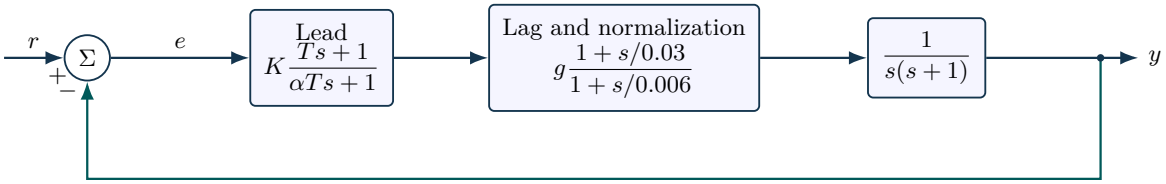
Coordinate and actuation
Positive angle is measured from the dashed reference ray. Torque acts about the marked gimbal pivot.

Normalized axis
$G(s) = \dfrac{1}{s(s+1)}$.
Lead supplies phase; lag changes low-frequency gain and the ramp error.

Limits of the view
Encoder and lens establish the signal meaning. Camera inertia, gearing and dimensions do not identify this stipulated plant.

A camera gimbal illustrates angular tracking and disturbance rejection. Its normalized plant is stipulated, with no claim that geometry alone determines its two poles.

From equipment to equations

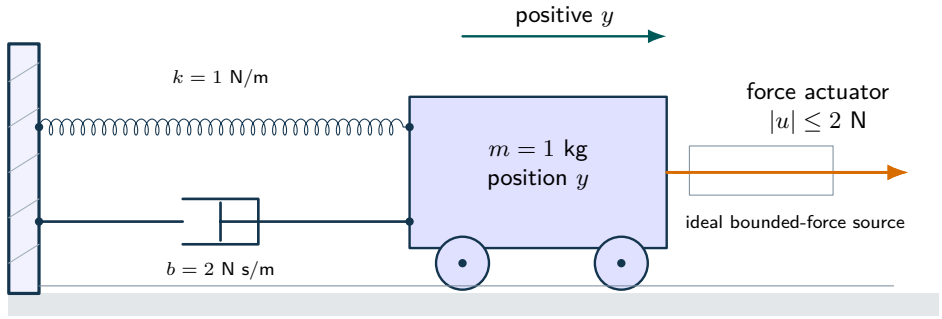




Frequency order is schematic, not a logarithmic scale.
The lag adds a small phase cost at the retained crossover.

Lead and lag sections are drawn separately so their gain normalization and frequency ordering can be analyzed independently.

09. PID regulation when the actuator saturates



Illustrative force-driven stage; no contact, friction or hard stops

Force and displacement

u: applied force [N].
y: displacement [m].
Requested and applied inputs differ when saturation is active.

Dynamics and bound

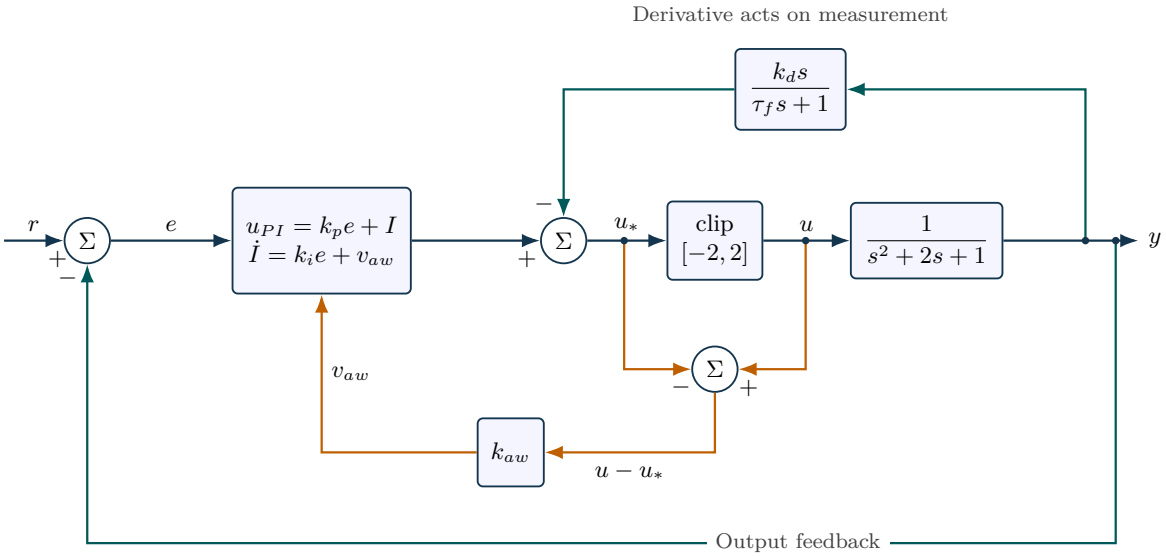
$\ddot{y} + 2\dot{y} + y = u$.
$-2 \leq u \leq 2$ N.
The drawn source applies force to the moving mass.

Controller implication

Anti-windup must use applied-minus-requested effort. Wheels depict ideal guidance; rolling friction and hard stops are omitted.

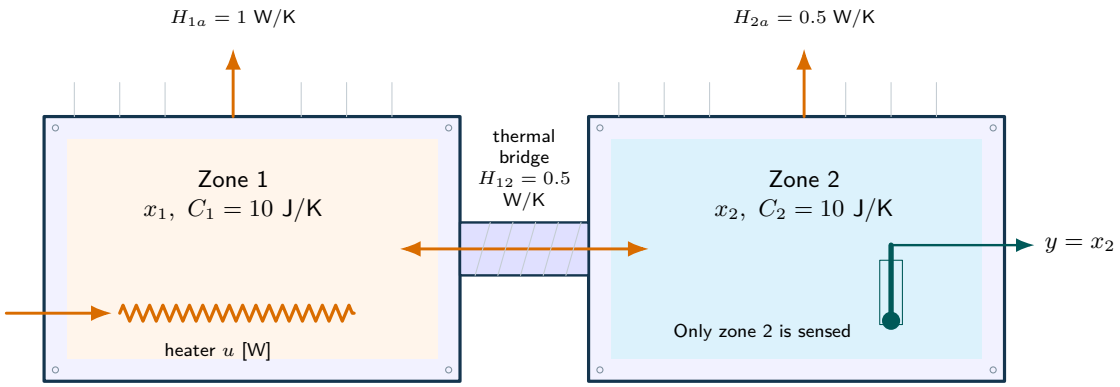
A mass, spring, damper and bounded force actuator physically realize the normalized second-order plant. Force saturation is included; contact, dry friction and travel stops are not.

From equipment to equations



Filtered derivative acts on measurement. Back-calculation sends the actual-minus-requested actuator effort into the integral-state equation.

10. Two thermal zones with one temperature measurement



Temperatures are deviations from a fixed ambient temperature

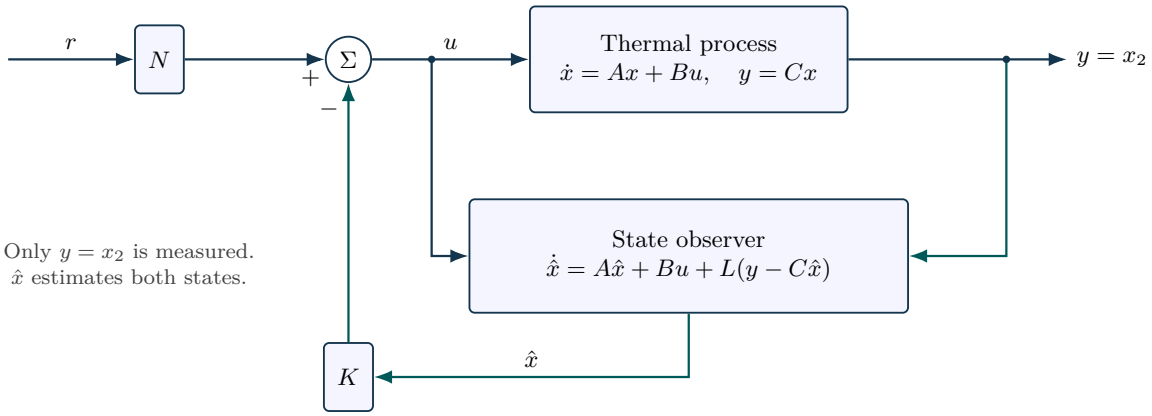
Two energy stores
$C_1 = C_2 = 10$ J/K.
x_1, x_2: ambient-relative temperatures.
Input heat enters zone 1 only.

Coupled balances
$10\dot{x}_1 = u - 1.5x_1 + 0.5x_2$.
$10\dot{x}_2 = 0.5x_1 - x_2$.
All right-hand terms are heat rates.

Sensing and estimation
$y = x_2$ supplies one measured coordinate. An observer reconstructs both states from the applied input and this output.

Two thermal inventories are connected by a conductive bridge. Heating acts in the first zone and sensing in the second. These stated capacities and conductances reproduce the two-state model exactly under the lumped assumptions.

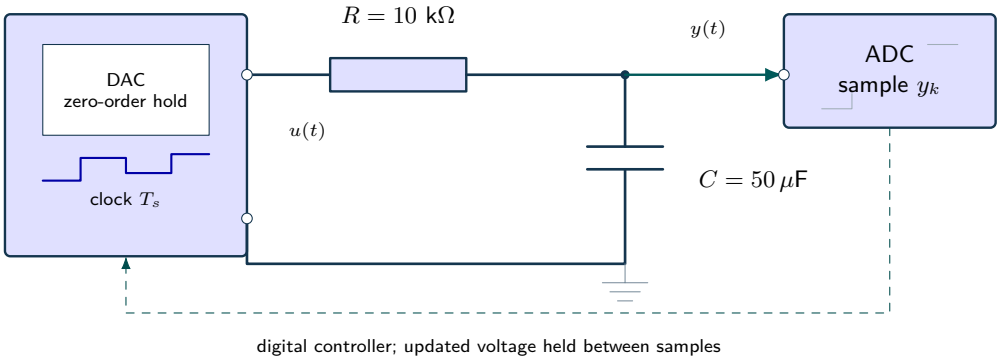
From equipment to equations



Controller: $u = Nr - K\hat{x}$
error: $e = x - \hat{x}, \quad \dot{e} = (A - LC)e$

The observer receives both the applied control and measured output. Its estimated vector supplies state feedback; the prefilter sets nominal reference gain.

11. Sampling can destabilize a continuous-time design



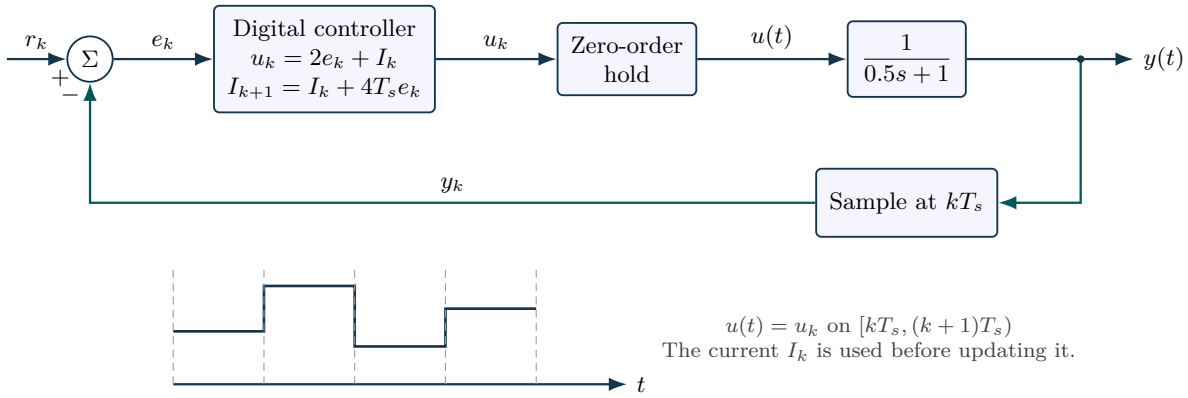
Continuous energy store
$RC = 0.5$ s; $\dot{y} = 2(u - y)$.
The held voltage drives a real-valued continuous state between samples.

Clocked interface
$y_k = y(kT_s)$.
$u(t) = u_k$ for $kT_s \le t < (k+1)T_s$.
Clock and update order matter.

Ideal converter boundary
No quantization, electrical loading or converter dynamics. Use exact held-input discretization when comparing sample periods.

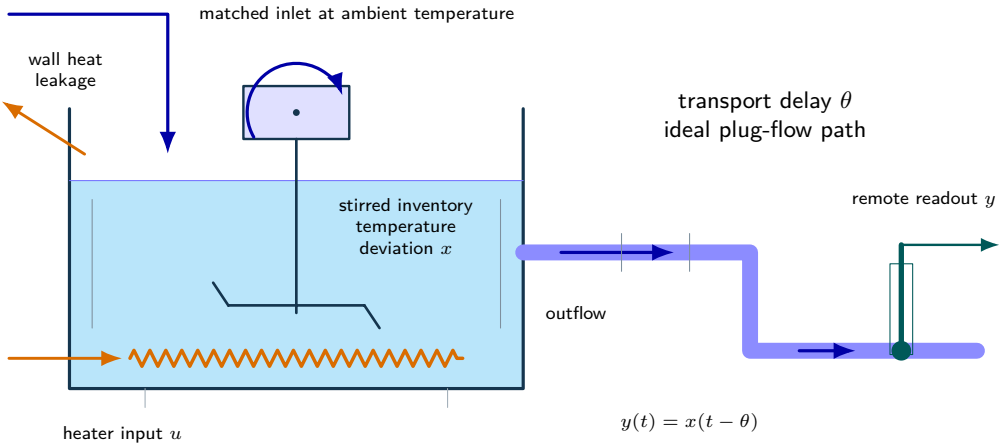
A held voltage drives an ideal RC network. The capacitor voltage is sampled by an ideal ADC; the software controller updates the DAC once per period. Quantization, converter lag and electrical loading are omitted.

From equipment to equations



The sampler, digital update order and hold are explicit. Continuous plant dynamics and sampled controller states occupy different parts of the loop.

12. A synthetic process from identification to validation



One physical interpretation of a first-order-plus-delay model

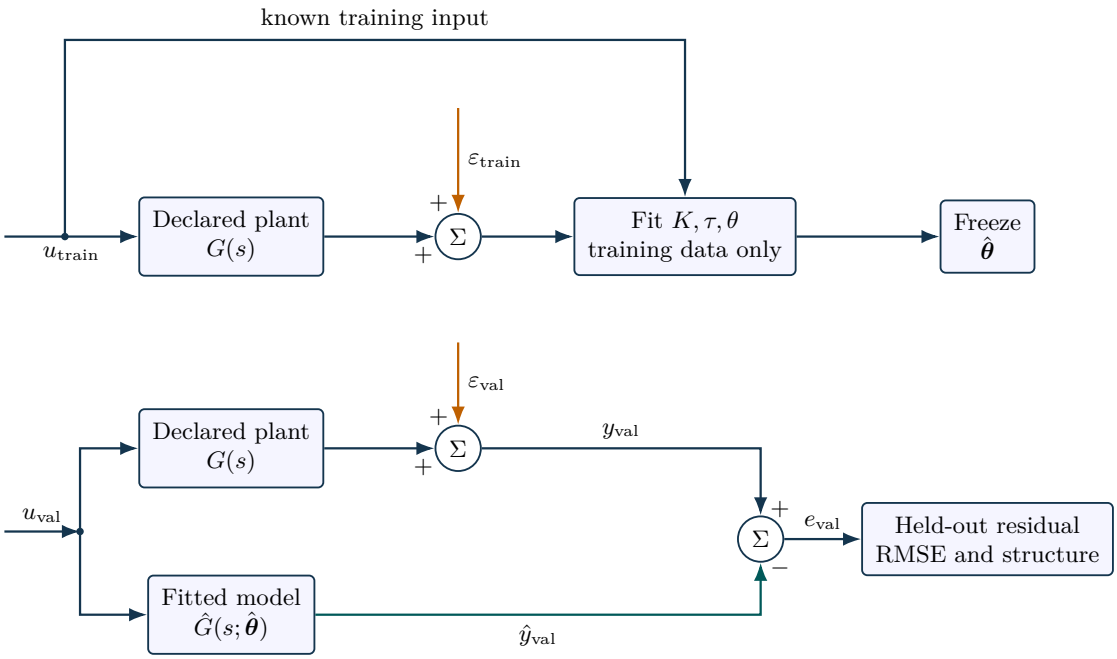
Synthetic experiment
$G(s) = 1.8e^{-0.6s}/(4s + 1)$. Declared gain, lag and delay generate the records; none is a physical measurement.

Thermal interpretation
One mixed energy store plus ideal transport. Through-flow and wall losses enter one effective conductance.

Identification boundary
Fit on training data, freeze the model, then check a separate multistep record. Geometry does not identify capacity or heater efficiency.

A stirred thermal inventory followed by ideal transport is one possible physical interpretation of the identified transfer function. The effective conductance includes wall leakage and through-flow heat removal. Its fitted gain and time constant do not separately determine capacity, conductance and heater efficiency.

From equipment to equations



Different input record and independent noise; no refitting on validation data.
All records in this exercise are synthetic.

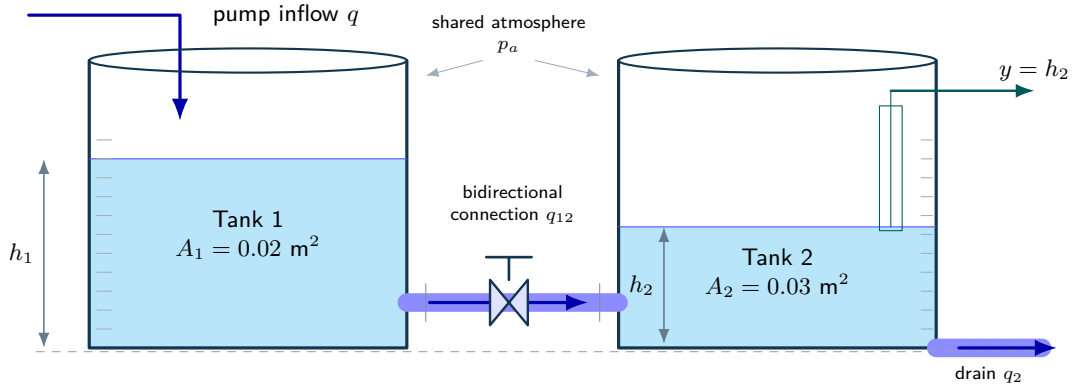
Training selects parameters; held-out validation freezes them. Separate noise records avoid reusing training residuals as validation evidence.

Integrated study: interacting water tanks

Two open tanks have constant areas $A_1 = 0.02$ m^2 and $A_2 = 0.03$ m^2. Their submerged connection obeys

$$q_{12} = k_{12}\,\mathrm{sgn}(h_1 - h_2)\sqrt{|h_1 - h_2|}, \qquad q_2 = k_2\sqrt{h_2},$$

where $k_{12} = 0.004$ and $k_2 = 0.003$ m$^{5/2}$/s. The pump feeds only tank 1. Nominal inflow is $\bar{q} = 0.0012$ m^3/s; the measured output is h_2. Areas and discharge coefficients are hypothetical. Assume nonnegative levels, no overflow and ideal sensing.



Common height datum; open tanks, constant areas, ideal gravity-flow laws

Two volume inventories

$A_1\dot{h}_1 = q - q_{12}$.
$A_2\dot{h}_2 = q_{12} - q_2$.
Only tank 2 drains to the exterior.

Gravity-driven coupling

$q_{12} = k_{12}\,\mathrm{sgn}(h_1 - h_2)$
$\cdot\sqrt{|h_1 - h_2|}$.
$q_2 = k_2\sqrt{h_2}$.

Interpret the arrows

Displayed levels give forward connection flow; reversal is allowed. The open rims share atmospheric pressure and one elevation datum.

Interacting tanks share a submerged connection at a common datum. Flow reverses if the level ordering reverses. Only tank 2 drains to the exterior; this is not a cascade with a freely falling inter-tank stream.

The six-part case derives conservation laws, local dynamics, finite-step error, PI stability, pump feasibility and a validation protocol.

Reuse in the five integrated studies

Case 1 uses the tanks on this page; case 2 uses laboratory 03; case 3 uses laboratory 05; case 4 combines the camera of laboratory 08 with the delay path of laboratory 07; case 5 combines the thermal plant of laboratory 10 with the held-input timing explained in laboratory 11. The RC plant is only the sampling example, not the plant of case 5.